

Annual review · Grade 8 geometry

The whole year on one page — and what to carry into Grade 9.

LESSON 68

1 × 40 MIN

GEOMETRY 8 – ANNUAL REVIEW

STAGE 9 · UNITS 3, 7, 13, 14

LESSON CLOCK

5'

14'

14'

7'

What do you remember?	5 min
The four chapters	14 min
Mixed problems	14 min
The summer plan	7 min

LEARNING OBJECTIVES

- Recall the four chapters of the year and their key theorems.
- Choose the right tool for an unfamiliar problem.
- Identify personal weak topics from the year’s control works.
- Set the summer work that follows.

TEXTBOOK

National	Темы 1–62, whole course
Cambridge	Learner’s Book, whole course
Workbook	Workbook, whole course

01

Explanation

The year in four chapters

QUARTER	CHAPTER	THE KEY RESULTS
I	Quadrilaterals	angle sum 360° ; the parallelogram tests; the midline theorem; Thales
II	Right triangles	Pythagoras; $\sin$, $\cos$, $\tan$; $\sin^2\alpha + \cos^2\alpha = 1$; solving a triangle
III	Coordinates, vectors, area	distance and midpoint; $AB = OB - OA$; $S = ah$, $S = \frac{1}{2}ah$, $S = \frac{1}{2}(a + b)h$
IV	The circle	tangent $\perp$ radius; the chord property; inscribed angle = $\frac{1}{2}$ central

THE THREE QUESTIONS THAT SOLVE MOST PROBLEMS

1. Where is the right triangle? — Pythagoras and trigonometry follow.

2. Where is the vertex? — in a circle question, that chooses the angle rule.

3. What do I know, and what do I want? — write both down before choosing a formula.

The summer plan

Look back at your four control works. Two topics will appear more than once in your lost marks — those are your summer topics, and nothing else is.

WHAT GRADE 9 WILL ASSUME YOU KNOW COLD

- Pythagoras, in both directions.
- The three trigonometric ratios, and which to choose.
- All the area formulas, without looking them up.
- The inscribed angle theorem and its corollaries.

Twenty minutes a week on those four is enough. Nothing else about the summer matters mathematically.

02 Worked examples

EXAMPLE 1 A rhombus has diagonals 16 and 12. Find its side, its area, its height and the angle between a side and the longer diagonal.

① *half-diagonals 8 and 6*

The diagonals bisect each other at right angles.

② $a = \sqrt{64 + 36} = 10$

Pythagoras — Quarter II.

③ $S = \frac{1}{2} \cdot 16 \cdot 12 = 96$

Quarter III.

④ $96 = 10h, h = 9.6$

⑤ $\tan \alpha = \frac{6}{8} = 0.75, \alpha \approx 36.9^\circ$

Trigonometry again.

ANSWER

$$a = 10, S = 96, h = 9.6, \alpha \approx 37^\circ$$

EXAMPLE A chord of 24 cm lies in a circle of radius 13 cm. Find its distance from the centre and the area of the triangle it makes with the centre.

① *half-chord 12*

The perpendicular from the centre bisects it — Quarter IV.

② $d = \sqrt{169 - 144} = 5 \text{ cm}$

Pythagoras — Quarter II.

③ $S = \frac{1}{2} \cdot 24 \cdot 5$

Area — Quarter III.

④ $= 60 \text{ cm}^2$

Three chapters in one problem.

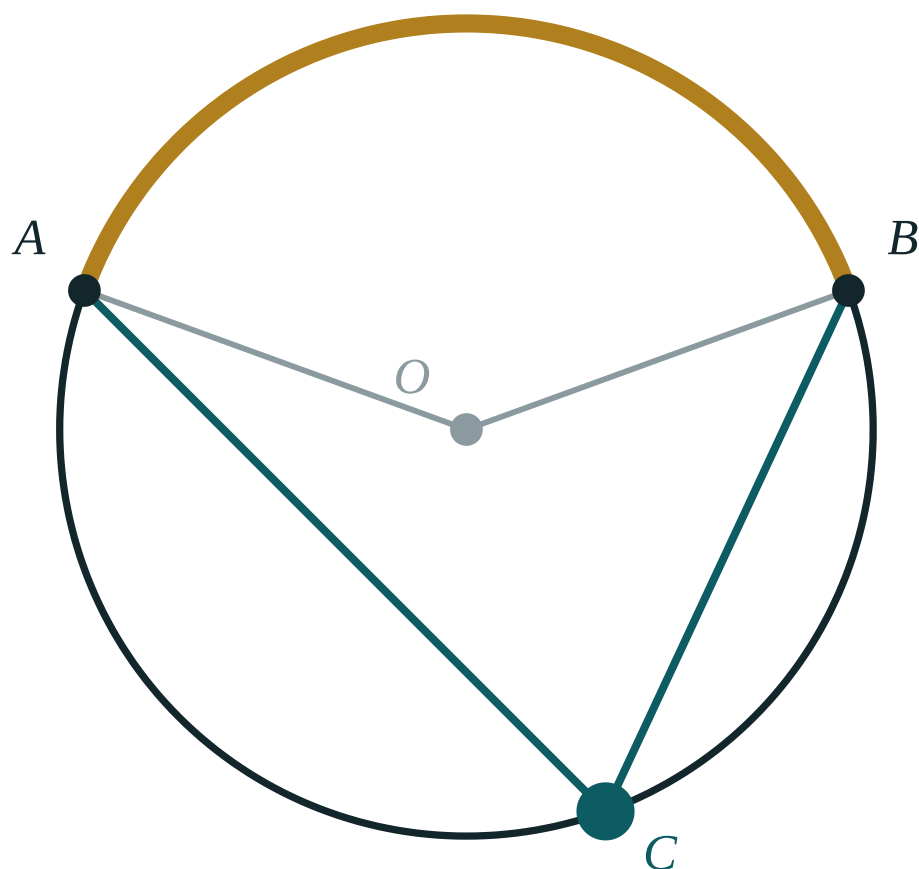
ANSWER

$d = 5 \text{ cm}, S = 60 \text{ cm}^2$

03 Interactive model

Revisit any model from the year — every one is still one click away from the lesson list.

A last look at the circle



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Annual review	Yillik takrorlash	Годовое повторение
Quadrilateral	To'rtburchak	Четырёхугольник
Midline	O'rta chiziq	Средняя линия
Pythagoras' theorem	Pifagor teoremasi	Теорема Пифагора
Trigonometric ratio	Trigonometrik nisbat	Тригонометрическое отношение
Vector	Vektor	Вектор
Area	Yuza	Площадь
Circle	Aylana	Окружность
Similar figures	O'xshash figuralar	Подобные фигуры

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The angle sum of a quadrilateral is:

$\sin^2\alpha + \cos^2\alpha$ equals:

The area of a trapezium is:

An inscribed angle on a 120° arc is:

120°

60°

240°

30°

AB in position vectors is:

$OA + OB$

$OB - OA$

$OA - OB$

$2OB$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find the angle sum of a hexagon.

ANSWER 720°

2. A right triangle has legs 9 and 12. Find its hypotenuse.

ANSWER 15

3. Find the area of a triangle with base 14 and height 6.

ANSWER 42

4. $r = 10$, $d = 6$. Find the chord.

ANSWER 16

5. Find the midpoint of (2; 4) and (10; 8).

ANSWER (6; 6)

6. A central angle is 80° . Find the inscribed angle.

ANSWER 40°

7. Find the area of a circle of radius 5 (in terms of π).

ANSWER 25π

Medium · the standard question

7 PROBLEMS

1. A parallelogram has sides 12 and 8 with an angle of 30° . Find its area.

ANSWER 48

2. A trapezium has parallel sides 8 and 14 and height 9. Find its area.

ANSWER 99

3. $OA = (1; 4)$, $OB = (7; 12)$. Find AB and $|AB|$.

ANSWER $(6; 8)$, 10

4. A ladder 5 m long leans with its foot 3 m from the wall. How high does it reach?

ANSWER 4 m

5. In a cyclic quadrilateral $\angle A = 88^\circ$. Find $\angle C$.

ANSWER 92°

6. A rhombus has diagonals 10 and 24. Find its side and its area.

ANSWER 13, 120

7. A tangent from P is 15 and $r = 8$. Find OP .

ANSWER 17

Hard · stretch and reason

7 PROBLEMS

1. An isosceles trapezium has parallel sides 12 and 20 and legs 5. Find its height, its area and its base angles.

ANSWER $h = 3$, area 48, base angle $\approx 36.9^\circ$

2. A chord of a circle of radius 15 subtends a central angle of 90° . Find its length and the area of the minor segment.

ANSWER chord $15\sqrt{2} \approx 21.2$, segment $\frac{225\pi}{4} - 112.5 \approx 64.2$

3. Show that $(0;0)$, $(6;0)$, $(8;4)$, $(2;4)$ is a parallelogram and find its area and its height.

ANSWER Diagonals share the midpoint $(4; 2)$; area 24, height 4

4. A right triangle has hypotenuse 26 and one leg 10. Find the other leg, the area and the smaller acute angle.

ANSWER 24, area 120, angle $\approx 22.6^\circ$

5. Two tangents from P touch a circle of radius 12 with $\angle APB = 60^\circ$. Find OP and PA .

ANSWER $OP = 24$, $PA = 12\sqrt{3} \approx 20.8$

6. A cylinder holds 1 litre with $h = 2r$. Find r , h and its surface area.

ANSWER $r \approx 5.4$ cm, $h \approx 10.8$ cm, $S \approx 554$ cm²

7. Name, for each quarter of the year, the single theorem you would keep if you could keep only one, and say why.

ANSWER A personal answer — but strong candidates are the parallelogram bisection test, Pythagoras, the area formula $S = \frac{1}{2}ah$, and the inscribed angle theorem, because every other result in its chapter can be rebuilt from it.

07 Homework

Summer work — 5 tasks

Twenty minutes a week keeps Grade 9 easy.

1. List the two topics that cost you most marks across the year's four control works.
2. Five problems a week from each of those two topics. No more.
3. *Practise Pythagoras in both directions until it needs no thought.*
4. *Learn every area formula of Quarter III by heart.*
5. *Learn the inscribed angle theorem and its three corollaries.*